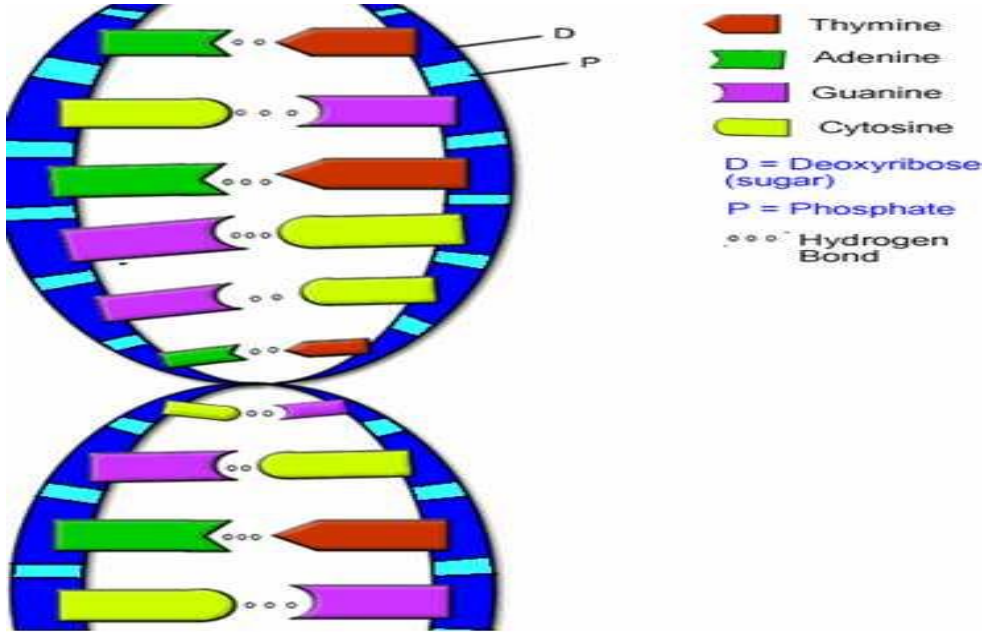
Lec 3- Genetic material and Bacterial replication

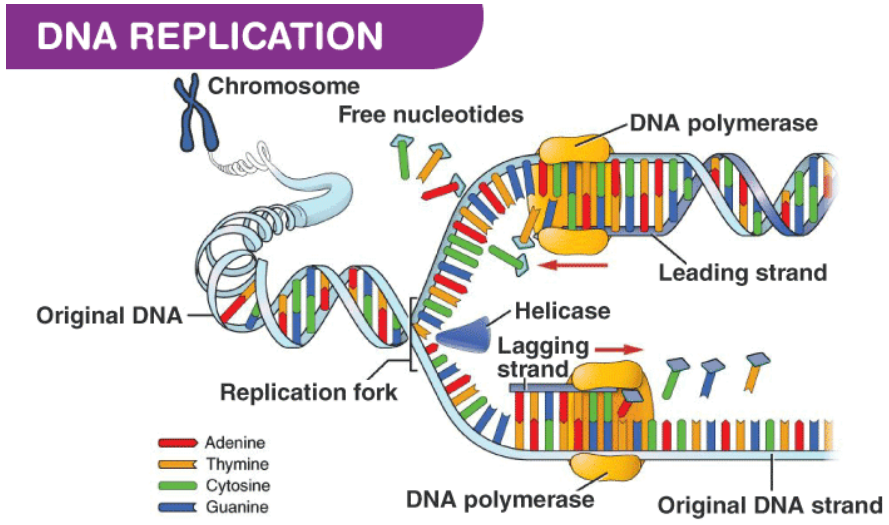


Prof. Dr. Marwa Salah

Professor of microbiology

Microbial Biotechnology Department

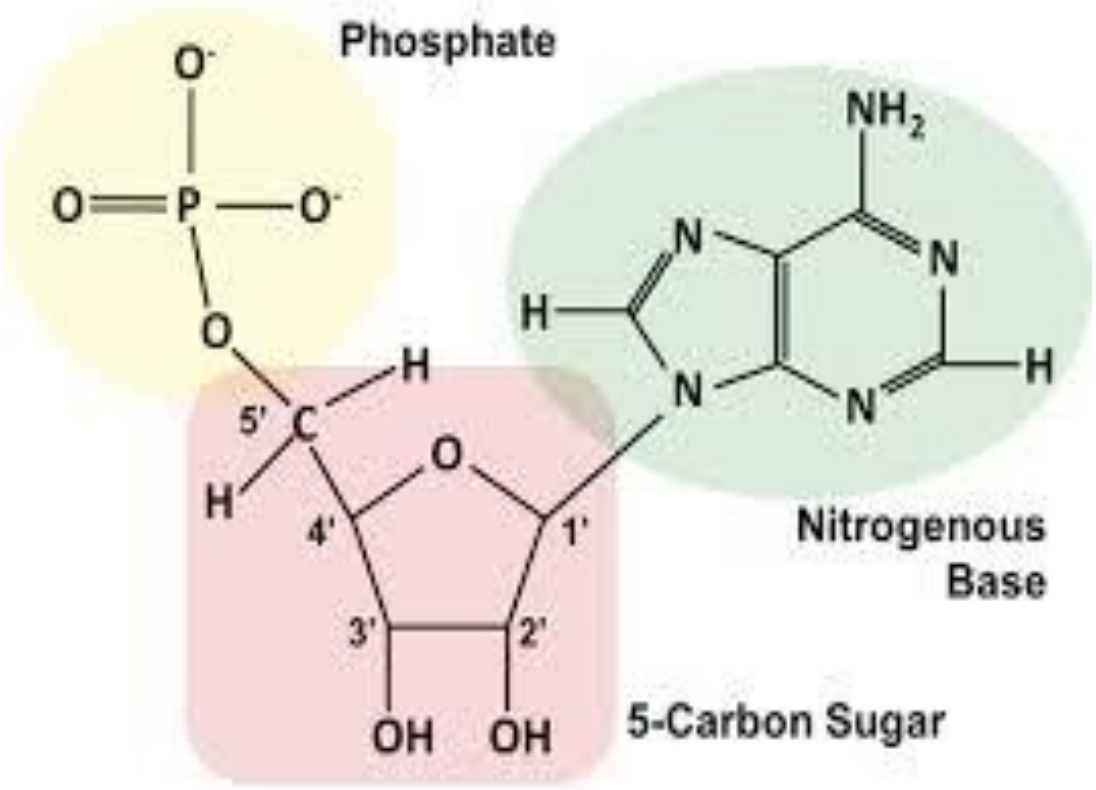
Faculty of Biotechnology

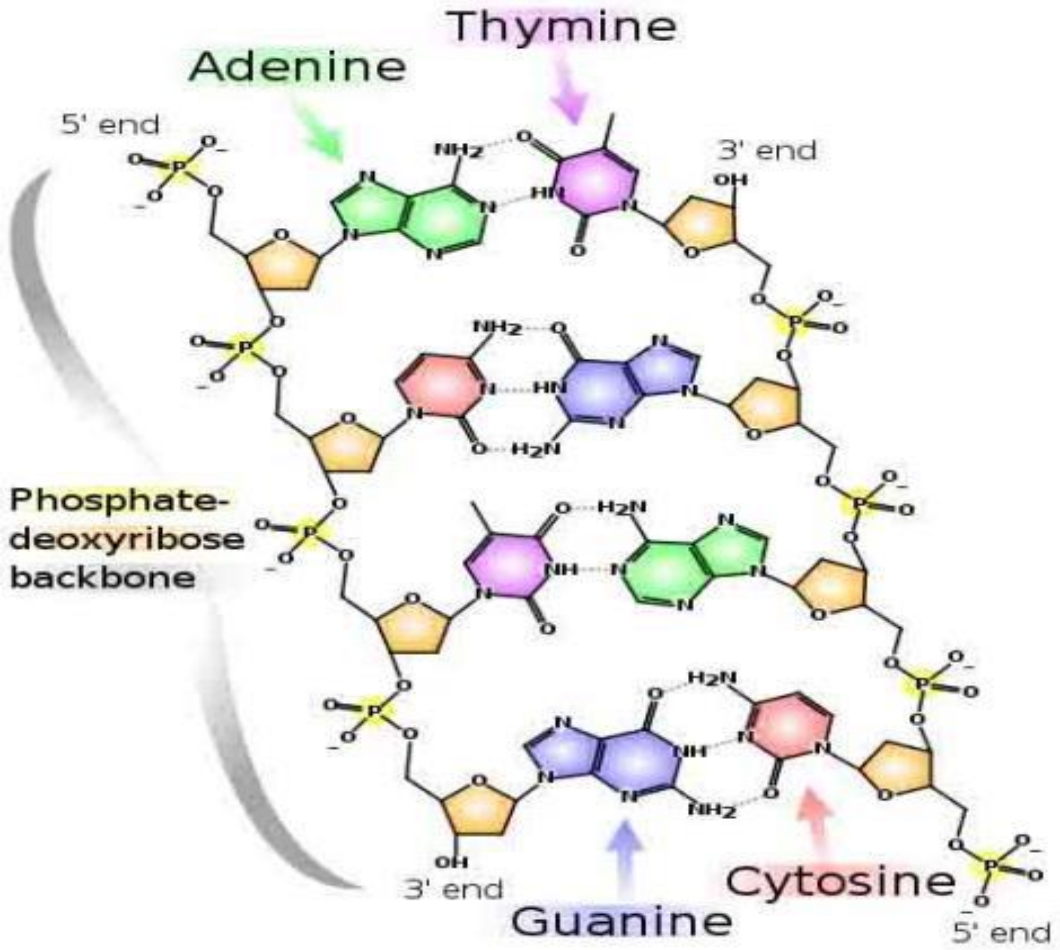


Microbial Genetics

Nucleic Acids: DNA Structure

Made of monomer building blocks called **nucleotides**.





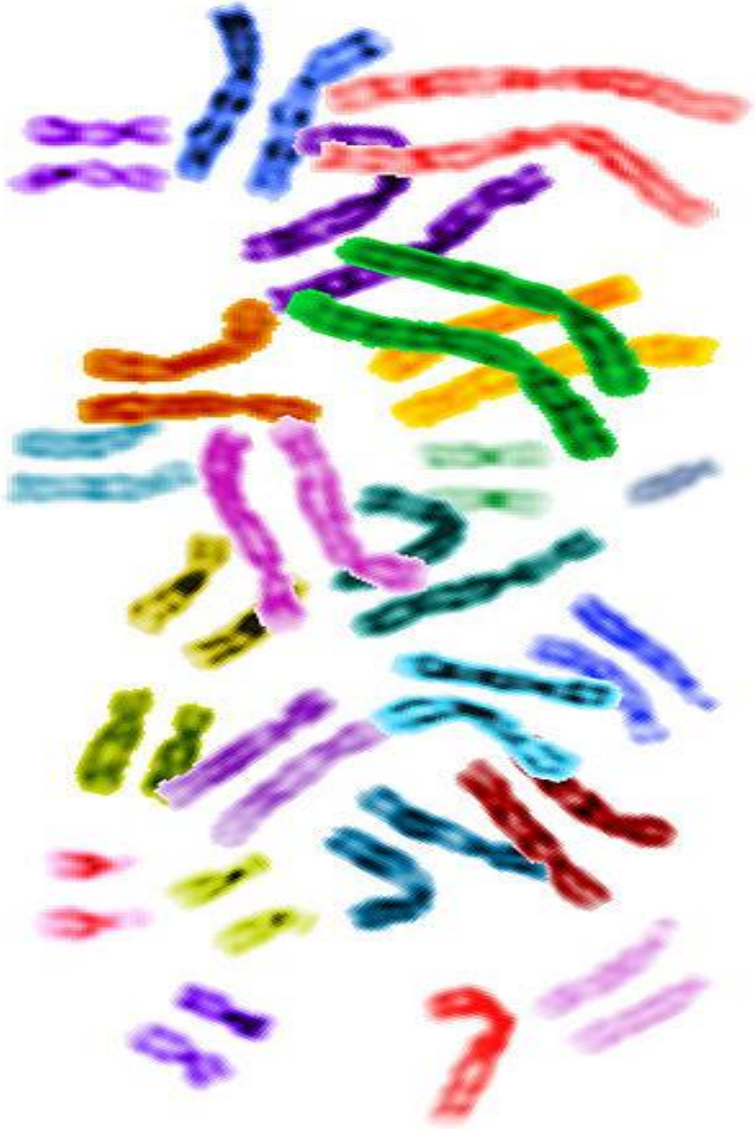
Eukaryote Genetics

- Like prokaryotes, insofar as genome is also made of DNA.

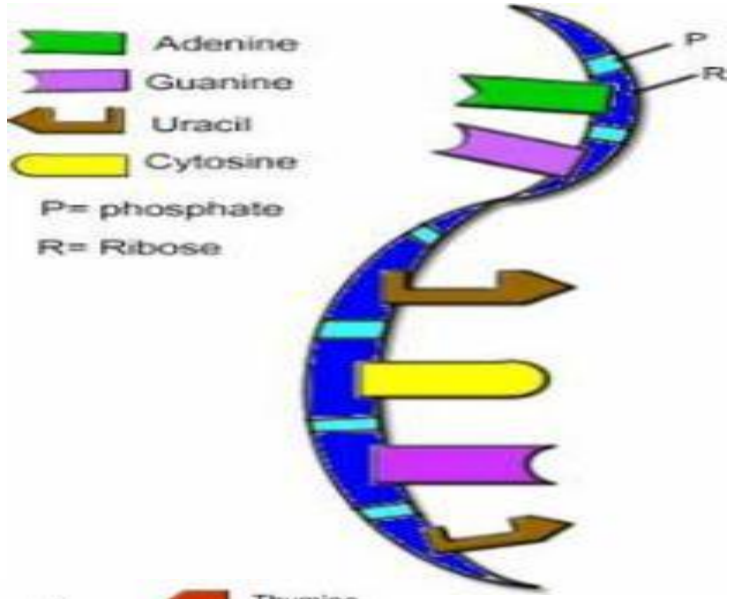
- May include several to many linear chromosomes within a membrane-bound nucleus.

- three locations of eukaryotic DNA
  - 1. Nuclear DNA
  - 2. Extranuclear DNA (mitochondria)
  - Plasmids in yeasts

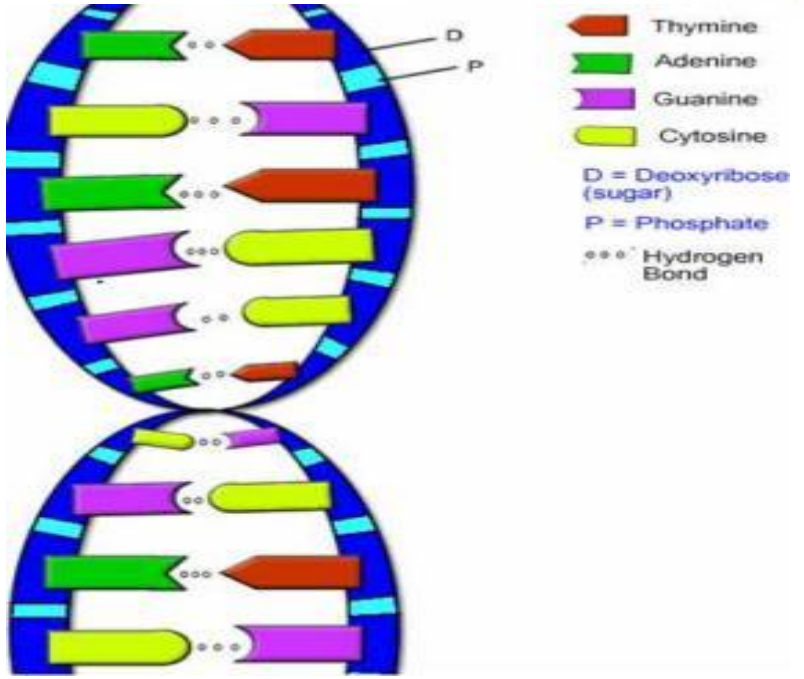
Q: Where is the extranuclear DNA located in eukaryotic cells?



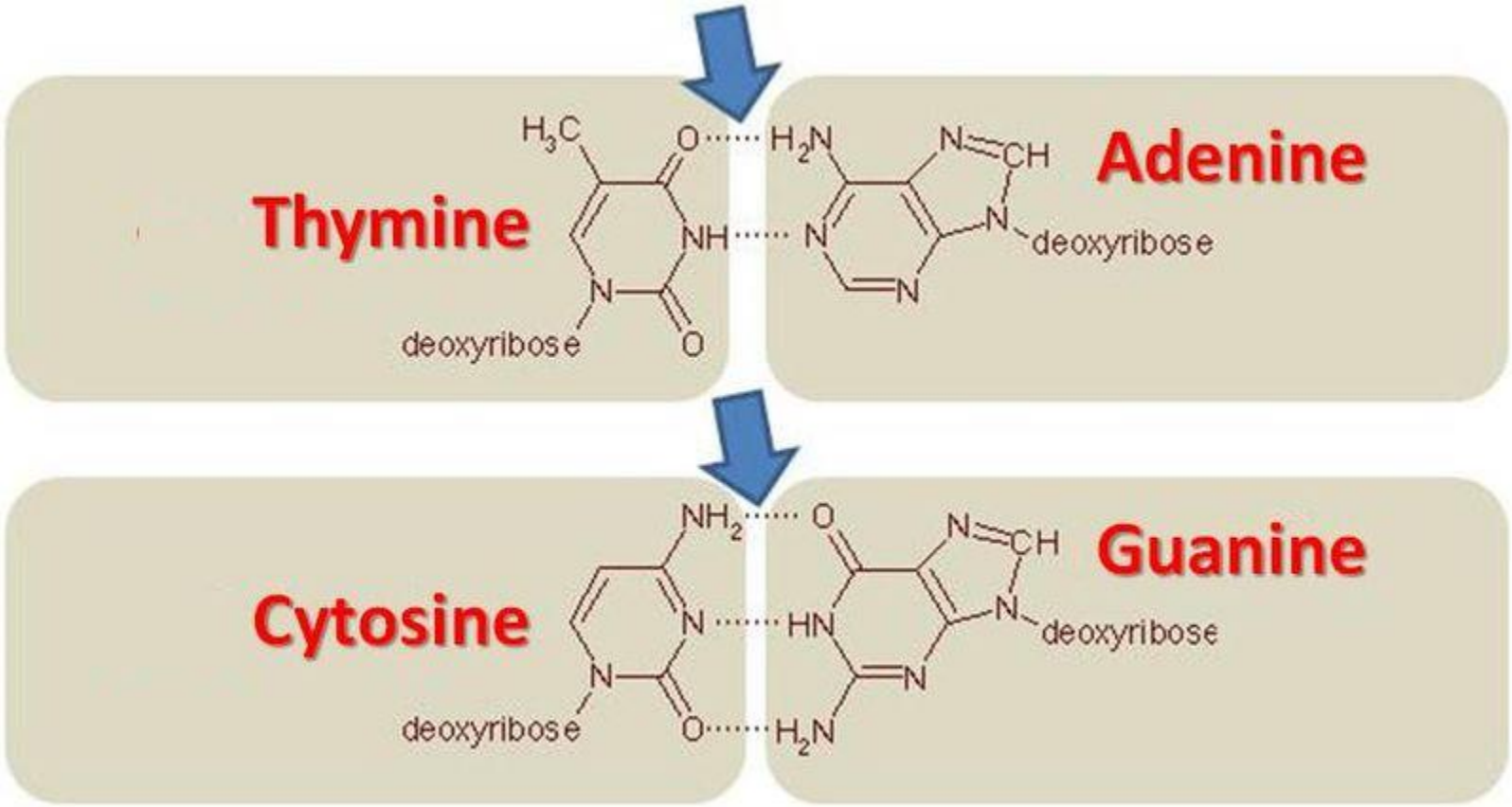
Genetic Material of Viruses



- Show more variety in nature of their genomes than do cells.
- Can be **DNA or RNA**; never both.
- Primary way scientists categorize and classify **viruses**.
- Can be dsDNA, ssDNA, dsRNA, ssRNA.
- May be linear and composed of several segments or single and circular.
- Much smaller than genomes of cells.



•No. of purines (double-ringed structures) present in DNA is exactly equal to the no. of pyrimidines (single ringed structures) present in the DNA. This is famously known as the **Chargaff's rule**.
•The bases stack above one another with the help of *Vander Waal forces* that act vertically between the bases.
•The major and minor grooves of the DNA are the primary sites for protein interaction.



the majority of DNA molecules exhibit certain chemical regularities that are defined as **Chargaff's base pairing rules** (base equivalence), namely the amount of A equals the amount of T and the amount of G equals the amount of C,

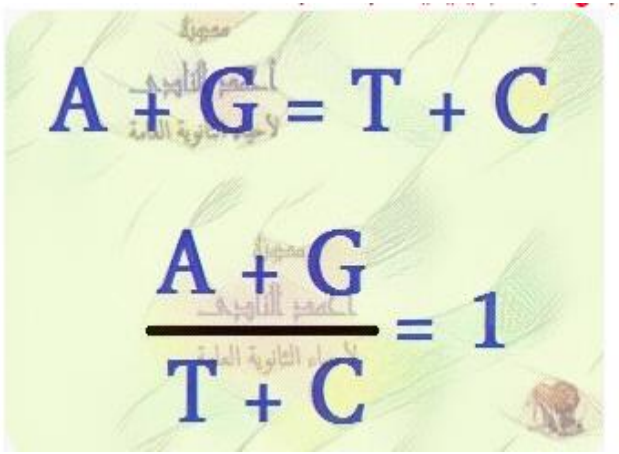
Determination [edit]

GC-content is usually expressed as a percentage value, but sometimes as a ratio (called **G+C ratio** or **GC-ratio**). GC-content percentage is calculated as[7]

$$\frac{G+C}{A+T+G+C} \times 100\%$$

whereas the AT/GC ratio is calculated as[8]

$$\frac{A+T}{G+C}.$$



40-60% GC content ensure stable binding of primer/template.

A higher GC-content level indicates **a relatively higher melting temperature.**

- Subsequent studies using space-filling models showed that there were **two grooves** in the DNA molecule, **one shallow** (approx. 12 Å) and **one deep** (approx. 22 Å).
- This feature was found to be important in relation to the binding of certain antibiotics, polymerase enzymes, etc.

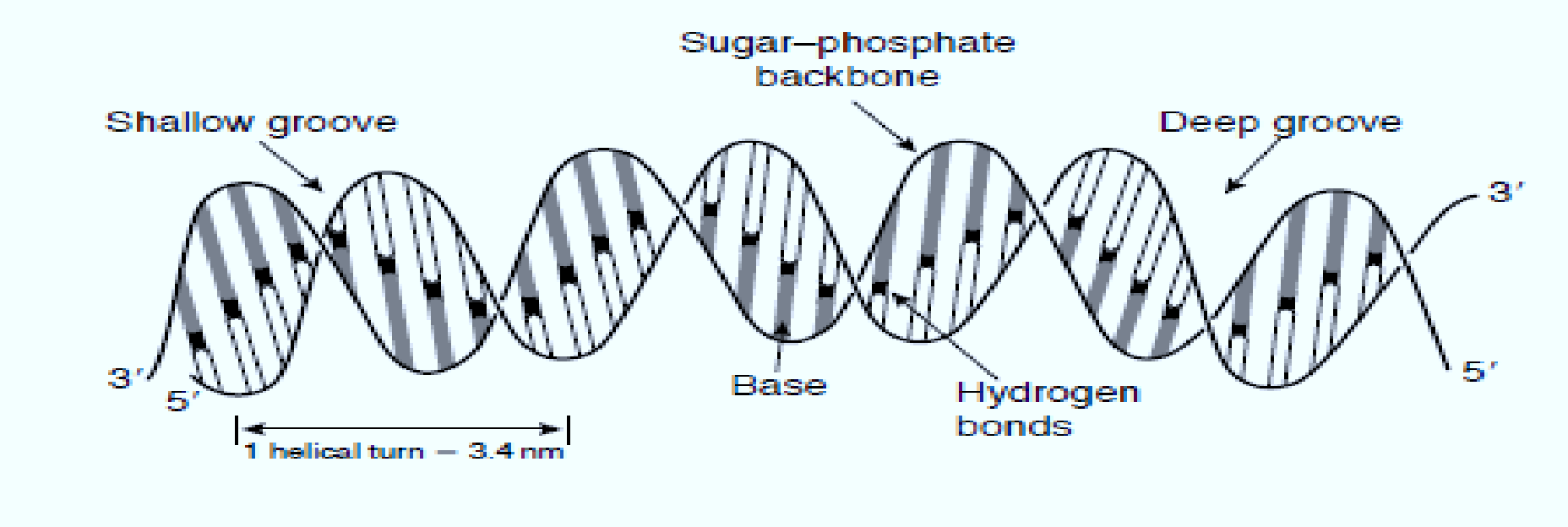


Table 3. DNA structure and function

Structural feature	Function
Strong, covalently bonded sugar–phosphate backbone	Preserves the base sequence of genetic information during the lifetime of the cell and allows shortening of the DNA during chromosome formation in cell division
Complementary purine–pyrimidine base pairing	Keeps an equal distance between the two polynucleotide strands, increasing the stability of the double helix
Numerous but weak H-bonds between complementary bases	Enhance stability of the helix but are easily broken by enzymes to allow DNA replication and transcription
Large, insoluble molecule (molecular mass approx. 100000 units)	Restricts DNA to the nucleus, protecting it from biochemical damage, thus preserving the genetic code

DNA replication

is semiconservative, meaning that each strand in the DNA double helix acts as a template for the synthesis of a new, complementary strand

- Process of copying DNA is called **replication**.
- Happens in all types of cells, **prokaryotes** and **eukaryotes**.
- **Q: Why would a cell need to copy its DNA?**

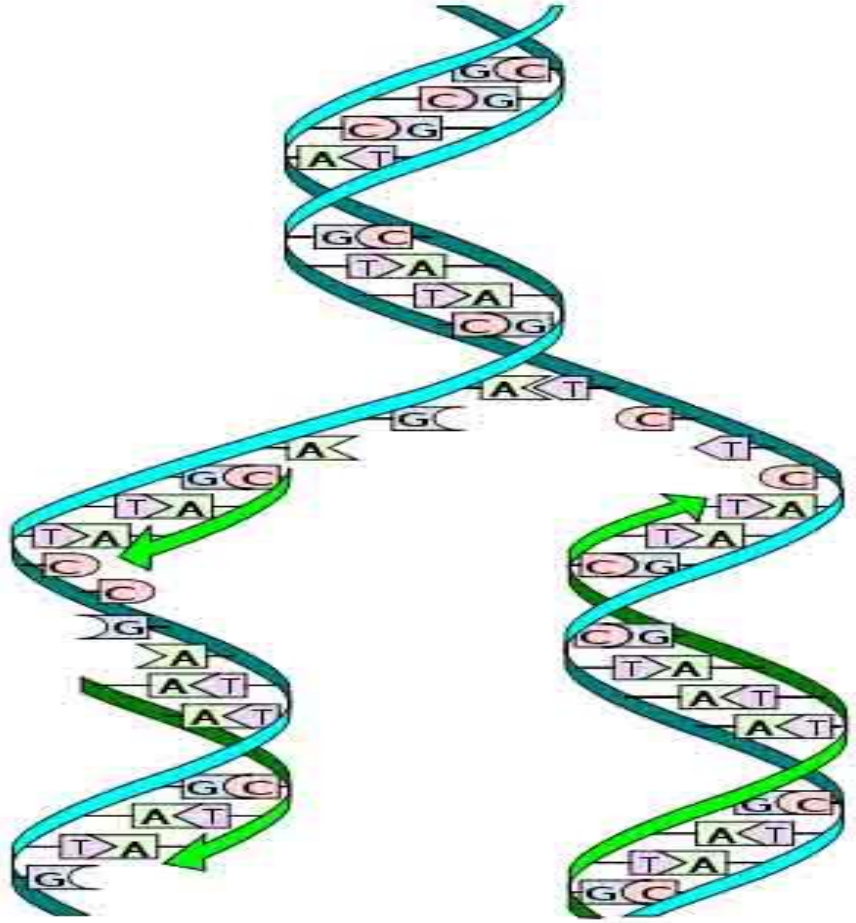
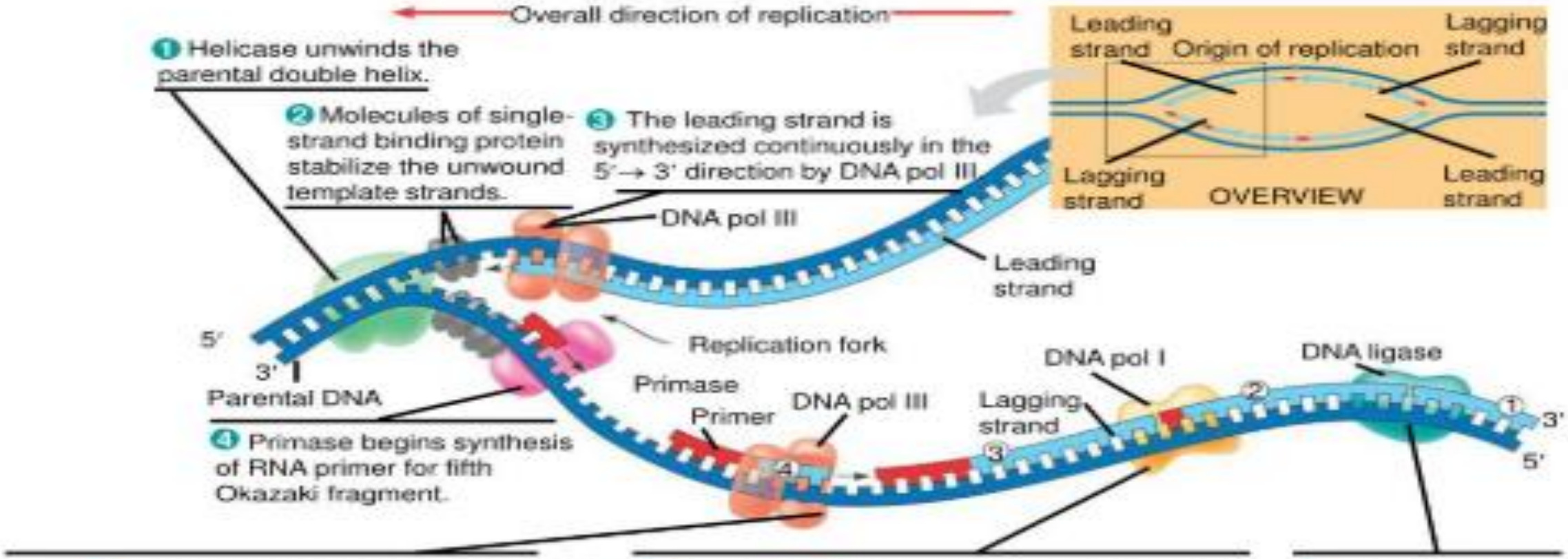


Figure 16.16 A summary of bacterial DNA replication



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Five steps of microbial DNA replication

Step 1: Identification of the origin of replication *(ori)*

It is the point where the DNA replication begins. Right next to ori there is an AT-rich portion of DNA to which ori binding proteins bind. The ori binding proteins unwind the DNA at ori. The unwound DNA is stabilized by the *single-strand binding proteins* (SSB) that prevent the reannealing of the DNA.

Step 2: The unwinding of the DNA

The further unwinding of the DNA is carried out by the enzyme helicase. As **the helicase enzyme** is acting, supercoiling occurs ahead of the replication fork and is taken care of by the enzyme topoisomerase.

Step 3: Formation of the replication fork

With the unwinding of the DNA, a replication **bubble is formed**. Each half of this replication bubble is referred to as a replication fork.

- The Okazaki fragments which are about **100-300bp in eukaryotes** are ligated together in a process known as Okazaki fragment maturation to complete DNA synthesis
- Okazaki fragments in bacteria and in bacteriophage T4 are **1000–2000 nucleotides long**
- Although there are well-defined replications **termination sequences** called **Ter sites** in prokaryotes, in eukaryotes, termination typically occurs by the collision of two replication forks.

Step 4: Actual DNA synthesis

Initially, an enzyme called **primase forms a 10-nucleotide long** RNA primer over which the **DNA polymerase III** adds further nucleotides. Later, the primer is removed by **DNA polymerase I**, and the gap thus created is also filled by it.

Step 5: Termination of DNA replication

After the DNA has replicated, specialized termination sequences of the DNA known as **ter** help in the termination of the DNA replication process. Ter sequence is bound by the tus proteins also known as **terminator utilization sequence**. Tus proteins act as **anti-helicase and stop the further replication of DNA**

S. No.	DNA Replication in Prokaryotes	DNA Replication in Eukaryotes
1.	Single origin of replication per DNA molecule	Multiple origin of replications per DNA molecule
2.	Occurs inside cytoplasm of the cell	Occurs inside nucleus of the cell
3.	It occurs only at one point	It occurs at multiple points
4.	It has one replication bubble	It has multiple replication bubbles
5.	It has one replicon	It has multiple replicons
6.	Initiation is carried out by DNA helicases	Initiation is carried out by multiple proteins like origin recognition complex (ORC), minichromosome maintenance (MCM) proteins, helicase loaders
7.	No licensing is required	Licensing for replication is required
8.	DNA polymerase I and III are involved	DNA polymerase α, ε, and δ are involved
9.	The rate of DNA replication is about 1000 bp per second	The rate of DNA replication is about 100 bp per second
10.	Okazaki fragments ranges from 1000-2000 bp long	Okazaki fragments ranges from 100-200 bp long

DNA Polymerase

- **Prokaryotes contain five different types of DNA polymerase**
- **It is the most important enzyme responsible for the replication of DNA.**
- **This enzyme complex reads the template DNA in the 3’ to 5’ direction and keeps adding the complementary base pairs in the 5’ to 3’ direction.**
- **This continuous addition of complementary base pairs produces a new daughter strand complementary to the template strand.**
- **The DNA polymerase III requires an RNA primer to start.**

- **DNA polymerase III is the main enzyme responsible for replication in prokaryotes.**
- **In eukaryotes, DNA polymerase δ is the main enzyme for replication.**
- **DNA polymerase I removes the RNA primer by 5'→3' exonuclease activity and replaces the primer by its polymerase activity in the lagging strand.**

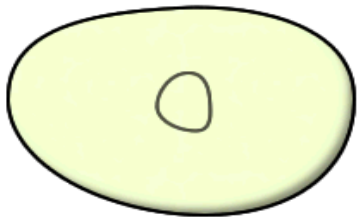
Main Features of Replication of DNA

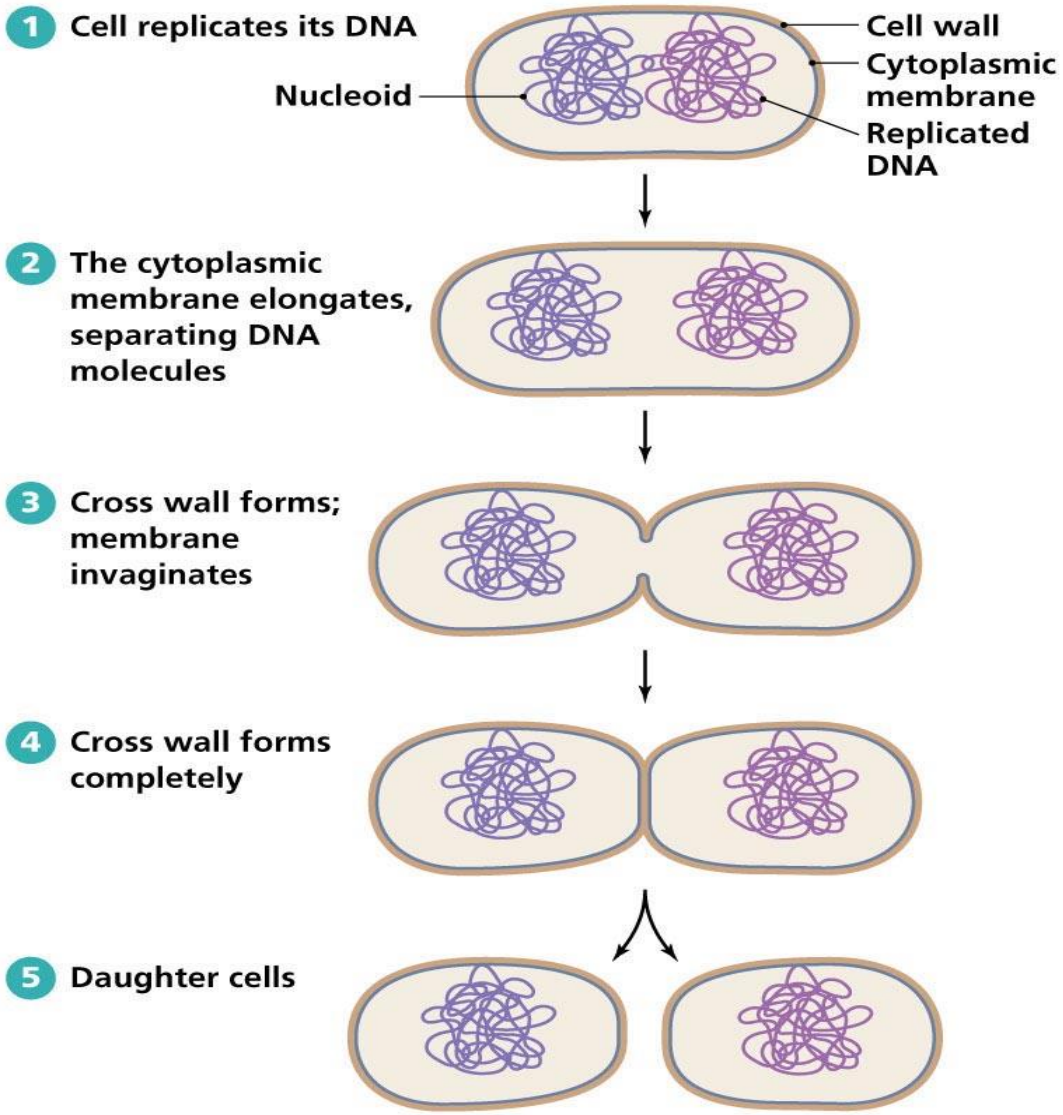
The role of replication in Genetics is to distribute all the genetic information equally between the two daughter cells.

- Both strands of the DNA act as a template.
- The direction of synthesis of a new strand is always from 5’ to 3’ direction. However, the reading of the template strand occurs from 3’ to 5’ direction.
- DNA replication is *semiconservative* in nature – one strand of the DNA is synthesized in a continuous technique which is known as the leading strand whereas the other strand is synthesized in a discontinuous mode, it is called the *lagging strand*.
- On the lagging strand, continuous reading of the template strand by DNA polymerase is not possible, hence the strand gets replicated in multiple small fragments known as *Okazaki fragments*. These Okazaki fragments are then joined together with the enzyme DNA ligase.

Prokaryote Genetics - Cell Division

After the genetic material is copied, the prokaryotic cell divides; a process called binary fission, in which two identical daughter cells arise from one parent cell.





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